Lab 4: Frequency-domain filtering

COMPUTER BASED SOUND PRODUCTION

One signal, three components

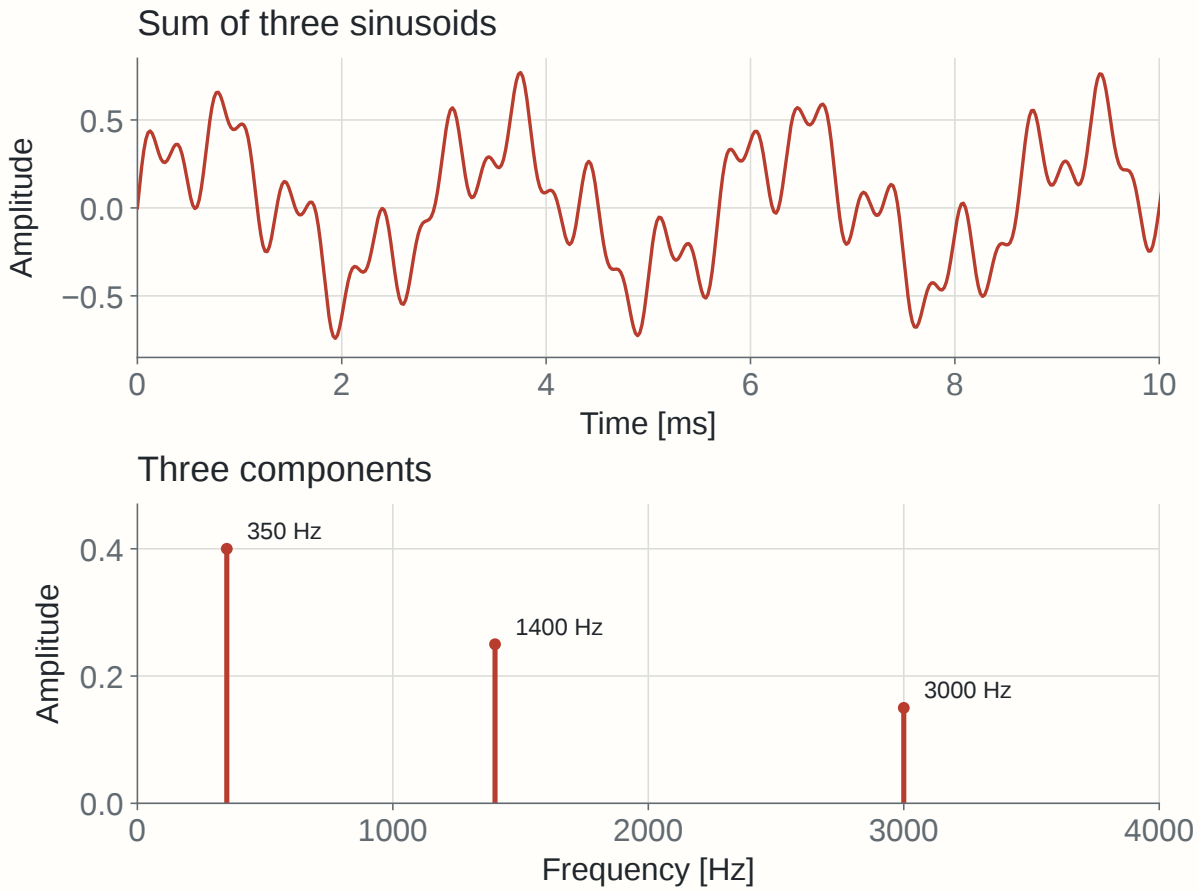
Our example is the sum of three sinusoids:

$$x[n] = \sum_{i=1}^{3} A_i \sin(2\pi f_i n / f_s)$$

n: sample index; $f_s = 48{,}000$ samples/s.

f_i	A_i
350 Hz	0.40
1400 Hz	0.25
3000 Hz	0.15

- The waveform combines all three components.
- Their positions use a linear frequency scale.



DFT and FFT

- The discrete Fourier transform (DFT) transforms N time-domain samples $x[n]$ into N complex frequency-domain coefficients $X[k]$, indexed by $k = 0, \ldots, N-1$.
- The fast Fourier transform (FFT) is a family of efficient algorithms for computing the DFT.
- The inverse DFT (IDFT) reconstructs samples from those coefficients; an inverse FFT computes it efficiently.

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \qquad k = 0, \ldots, N-1$$

$j^2 = -1$. A library normally handles the transform; understand its scaling and data layout.

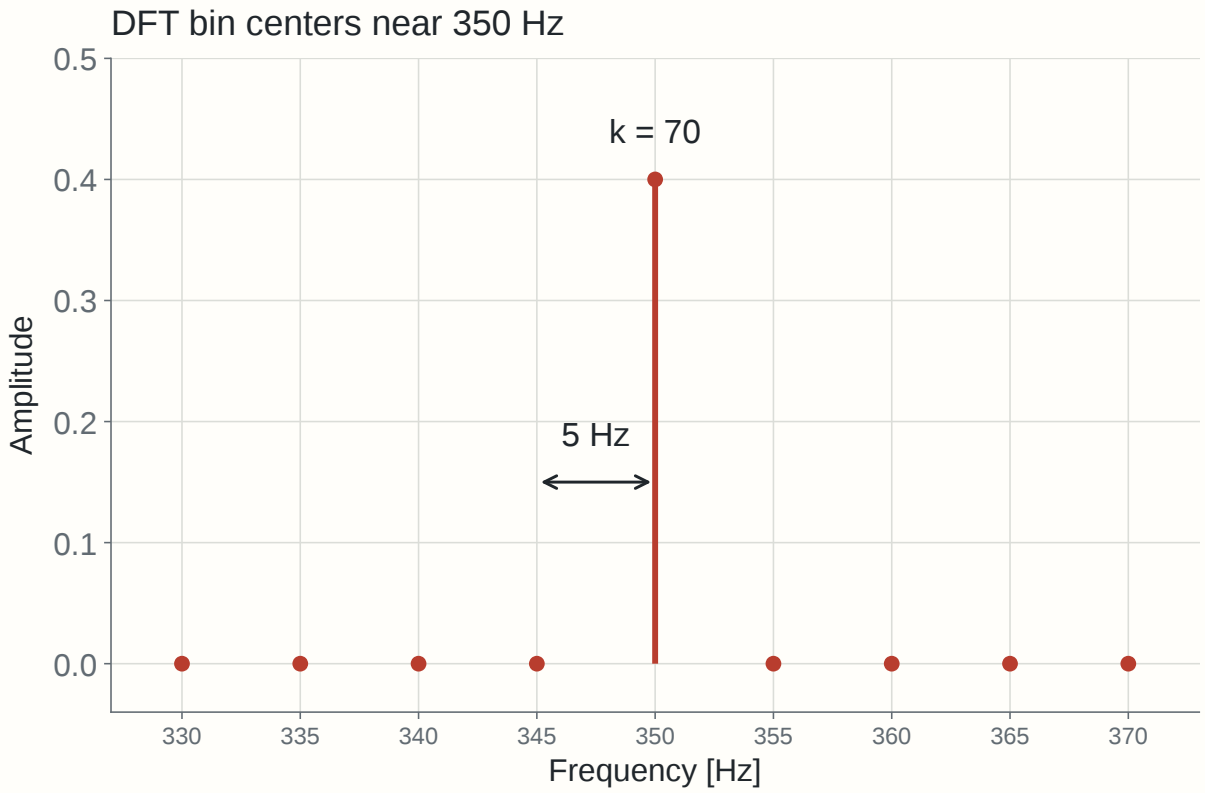
Frequency bins

For a block of N samples:

$$T = N/f_s, \qquad \Delta f = f_s/N = 1/T$$

Positive-frequency bins have centers $f_k = k\Delta f$.

- $f_s = 48{,}000$ Hz, $N = 9600$: $T = 0.2$ s, $\Delta f = 5$ Hz; 350 Hz is at $k = 70$.
- Even N: one-sided bins $k = 0, \ldots, N/2$; the last is at $f_s/2$ (Nyquist).
- Odd N: one-sided bins $k = 0, \ldots, (N-1)/2$; there is no Nyquist bin.
- Longer observation improves frequency discrimination.

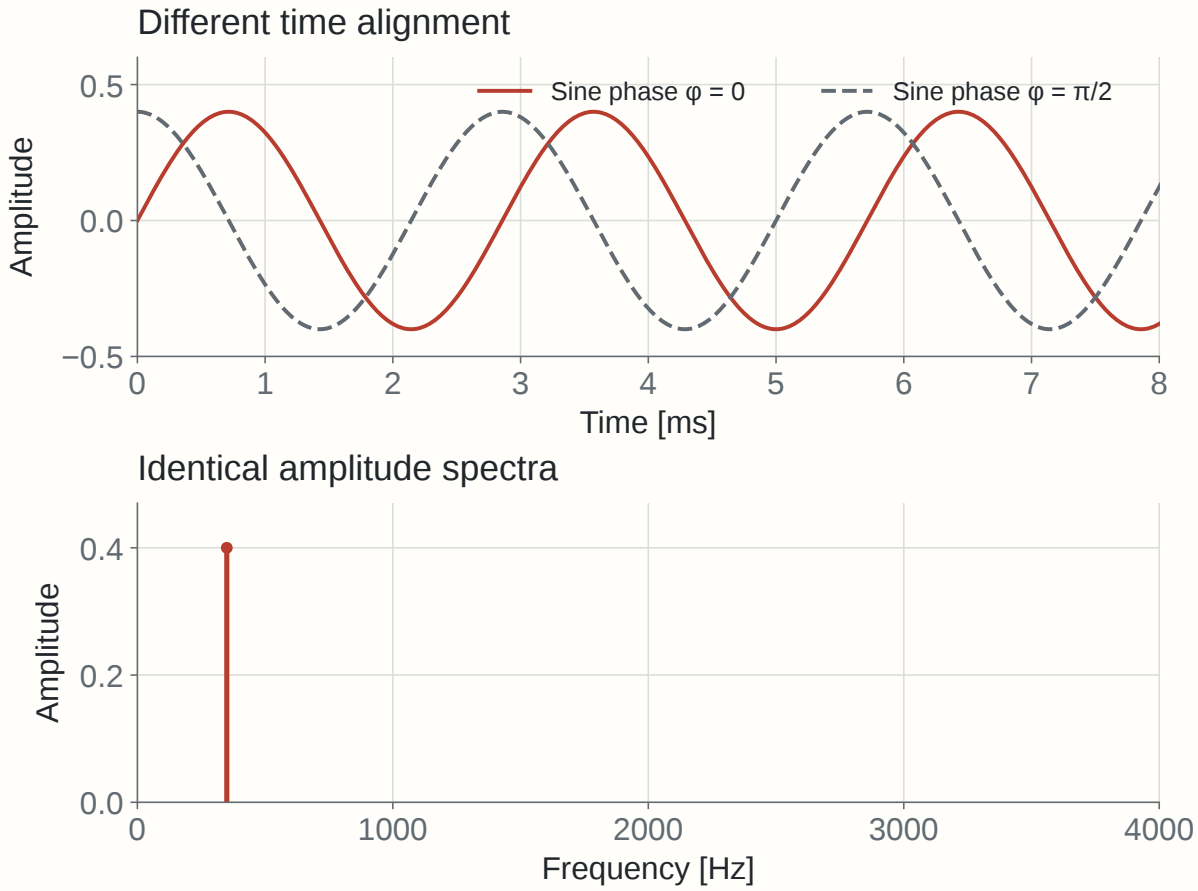


Magnitude and phase

Each coefficient can be written as:

$$X[k] = |X[k]|e^{j \arg X[k]}$$

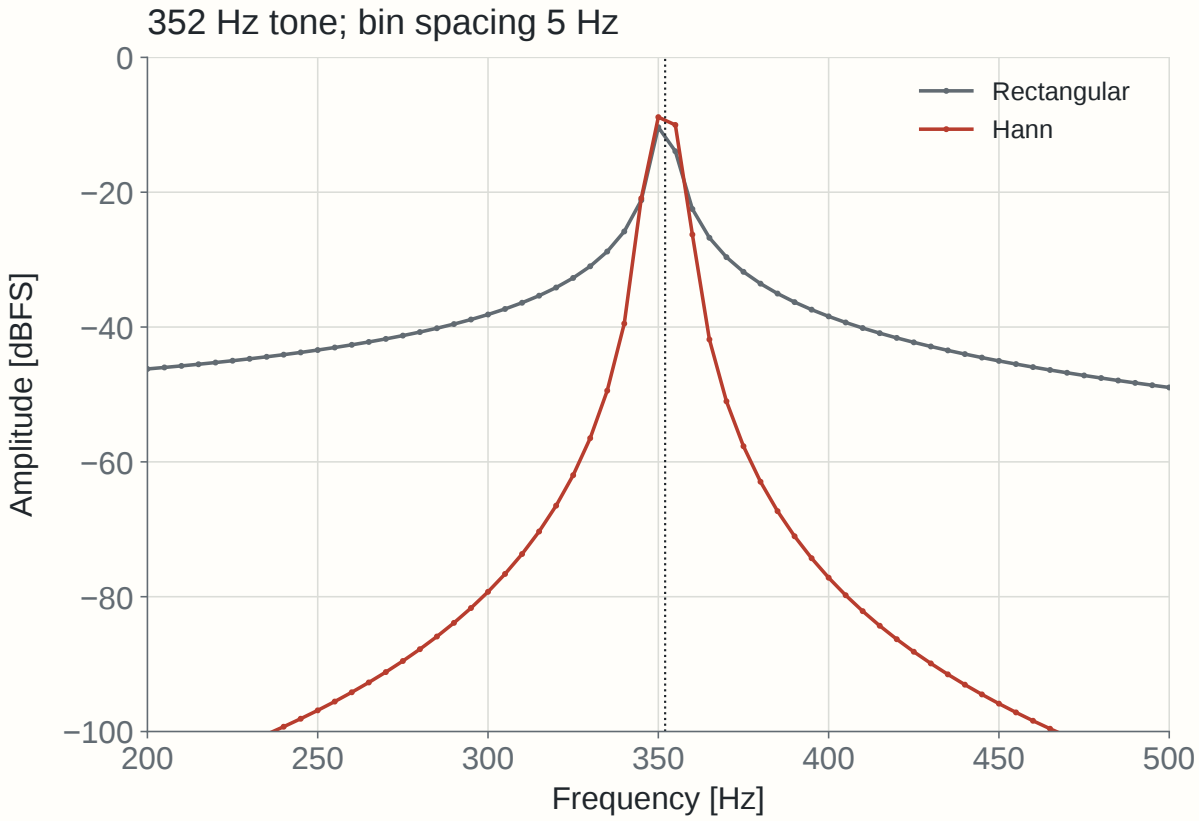
- Magnitude $|X[k]|$ describes the strength of a component.
- Phase $\arg X[k]$ describes its alignment in time.
- Two signals can have the same magnitude spectrum and different waveforms.
- Keep the complex coefficients when processing audio; magnitude alone cannot reconstruct the signal.



Finite blocks and spectral leakage

- The DFT treats the block as one period of a repeating signal.
- A tone between bin centers spreads across multiple bins: spectral leakage.
- An analysis window, such as Hann, reduces distant leakage but broadens the main lobe.
- Windowing changes amplitude scaling; account for its gain.
- Zero padding adds display points, not new information or a longer observation.

An isolated tone need not appear as one spectral line in a measured FFT.

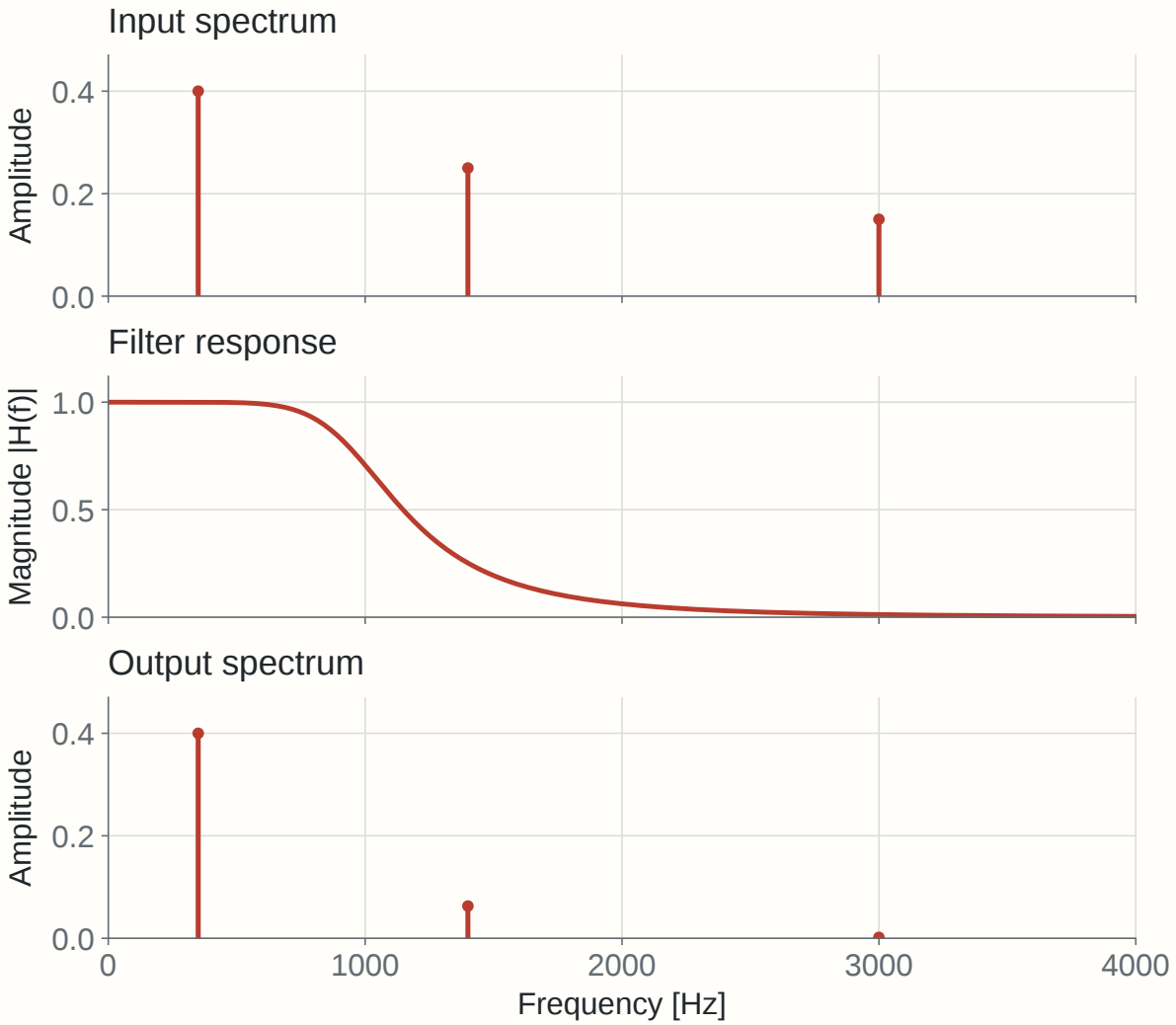


A filter's frequency response

- $X[k]$: spectrum of the input signal.
- $H[k]$: frequency response of the filter.
- $Y[k] = H[k]X[k]$: spectrum of the output.

The response can change magnitude and phase.

- A filter may attenuate or amplify components.
- A frequency-response graph describes behavior; it does not specify how the filter is implemented.
- Filters can operate directly on time samples or through FFT-based processing.



Why filter using the FFT?

Goal: reduce an unwanted 440 Hz tone while preserving surrounding sound.

A frequency response describes what changes; the implementation determines how.

Time-domain notch filter	Frequency-domain filtering
Compute each output sample from inputs and filter state.	Transform a block, multiply complex bins by a mask, then invert.
Efficient for a simple notch; little buffering.	Choose bins directly: reject 435–445 Hz in an offline recording.
Rejection width and shape follow the chosen filter design.	FFT convolution can also make long FIR filters efficient.

$$x[n] \longrightarrow X[k] \longrightarrow H[k]X[k] \longrightarrow y[n]$$

For a simple 440 Hz notch, a biquad is usually cheaper.

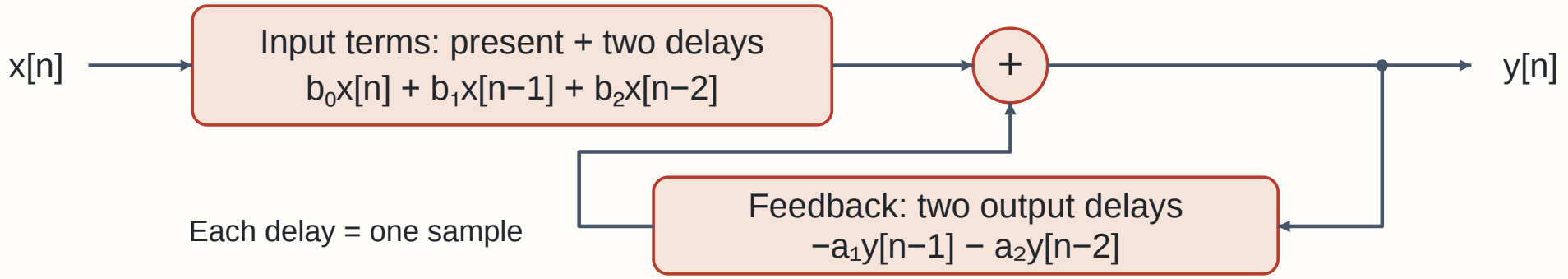
FFT filtering offers explicit spectral control, with buffering, boundary handling and possible ringing.

Time-domain filtering: the biquad

A second-order IIR (infinite impulse response) filter uses present and past samples:

$$y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2] - a_1 y[n-1] - a_2 y[n-2]$$

- Feedforward: weighted input samples; feedback: weighted previous outputs.
- Direct form I stores two previous inputs and two previous outputs; z^{-1} is a one-sample delay.
- Preserve state across blocks; each cascade stage has its own memories.
- Zero initial state can cause a startup transient; a stable filter may retain a decay tail.

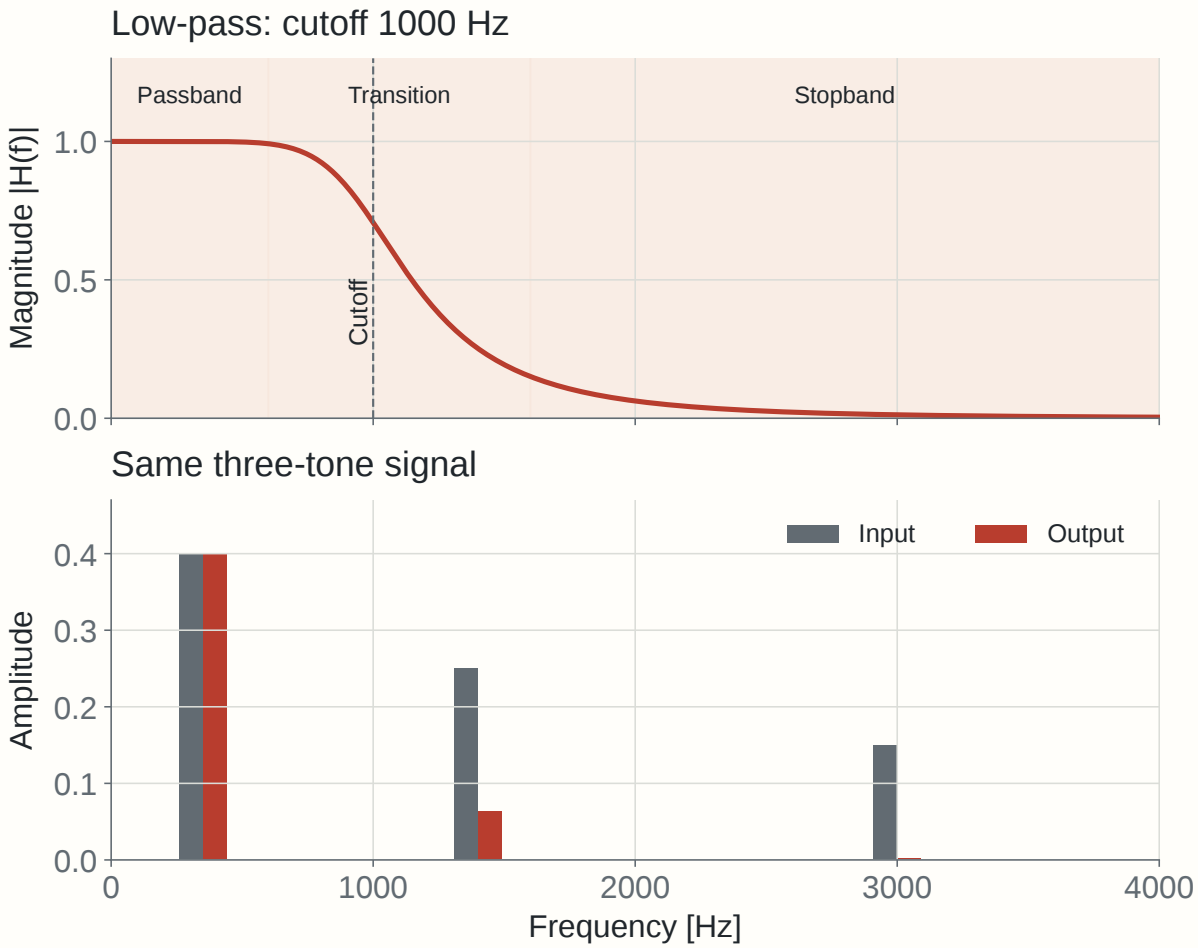


Low-pass filter

- Passes low frequencies and attenuates high frequencies.
- The cutoff frequency f_c marks a specified response level.
- A practical response has a transition band.
- Passband amplitude may be approximately constant; phase can still change.

Here: $f_c = 1000$ Hz at approximately −3 dB.

350 Hz is retained; 1400 and 3000 Hz are attenuated.

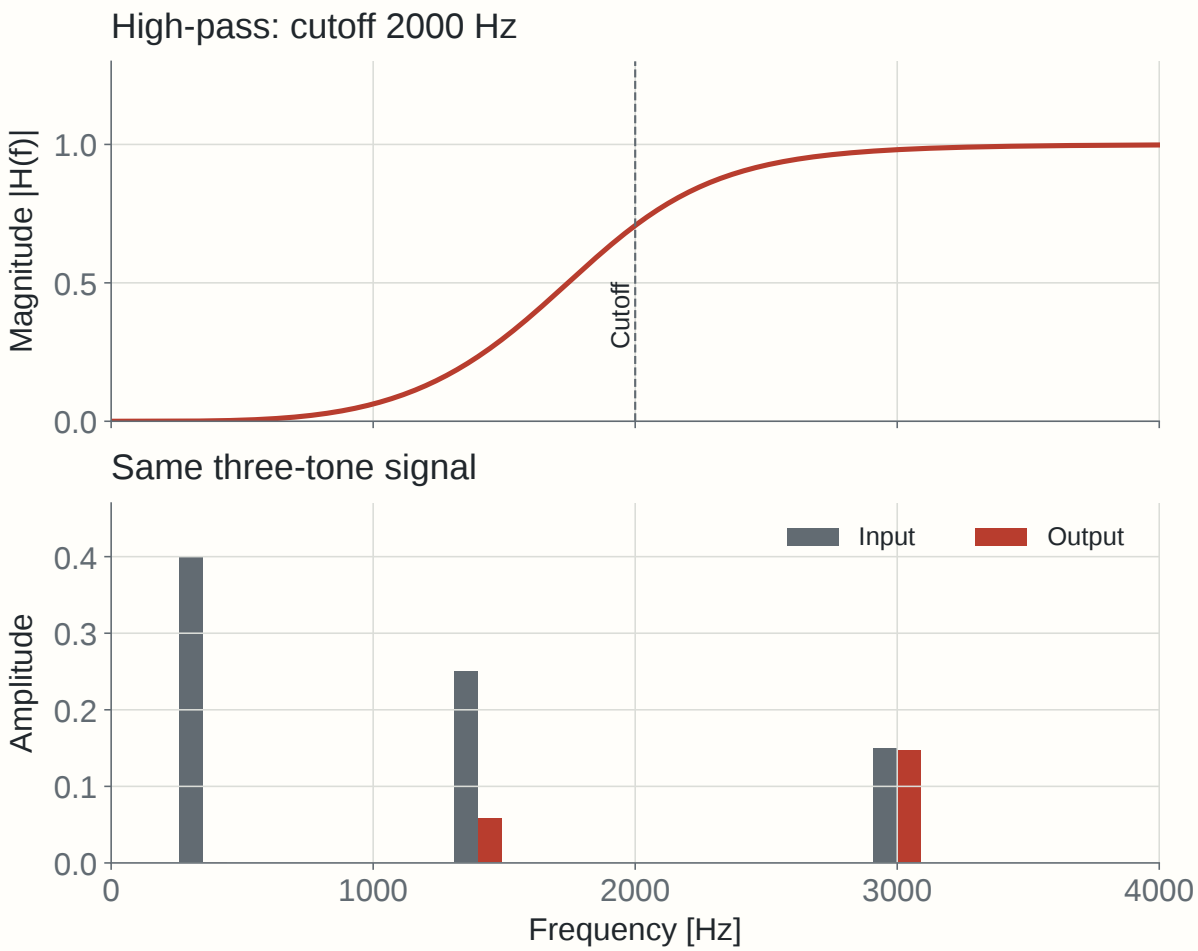


High-pass filter

- Attenuates low frequencies and passes high frequencies.
- The cutoff uses a stated response convention.
- The transition is gradual in this example.
- The DC component at 0 Hz is removed.

Here: $f_c = 2000$ Hz at approximately −3 dB.

3000 Hz is retained; 350 and 1400 Hz are attenuated.



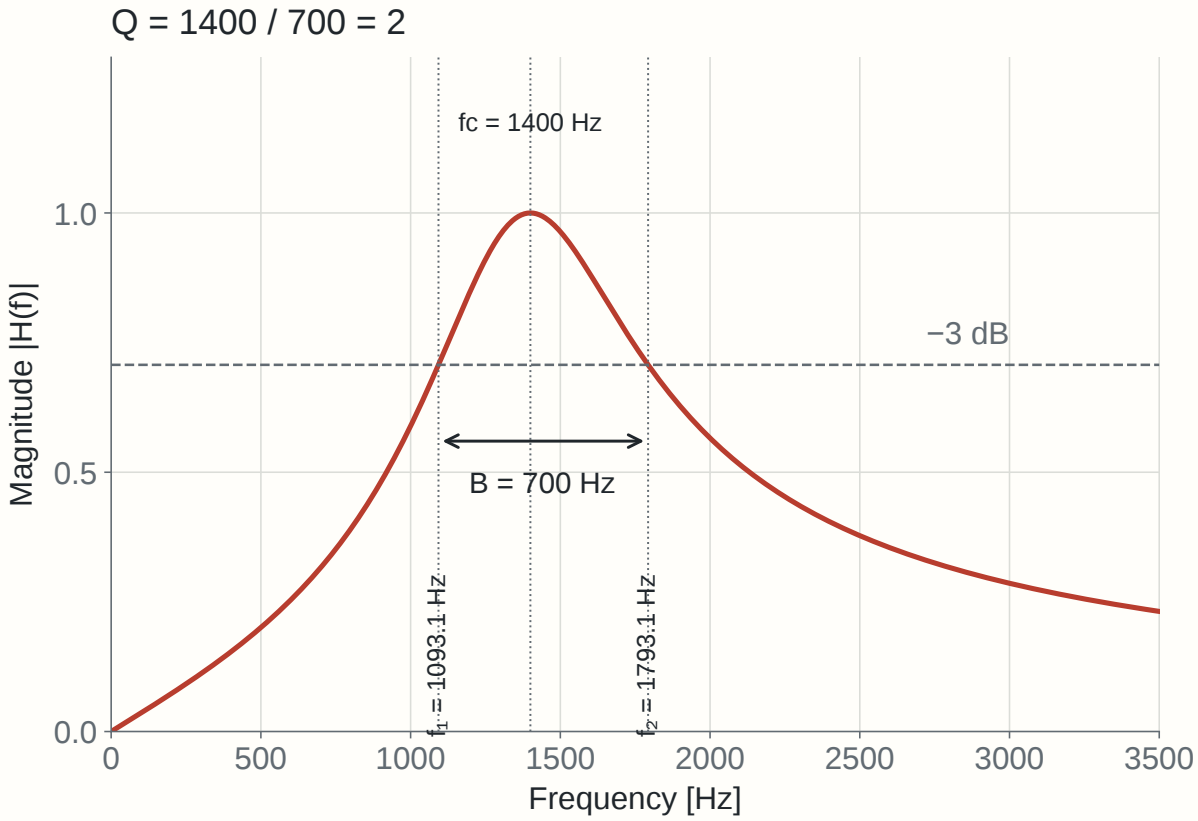
Bandwidth and quality factor

For the illustrated band-pass prototype:

$$B = f_2 - f_1, \qquad Q = \frac{f_c}{B}$$

- f_1, f_2: −3 dB points relative to the peak.
- At fixed f_c, a higher Q means a narrower band.
- Q is dimensionless; bandwidth B is in Hz.
- Example: $f_c = 1400$ Hz, $B = 700$ Hz gives $Q = 2$.

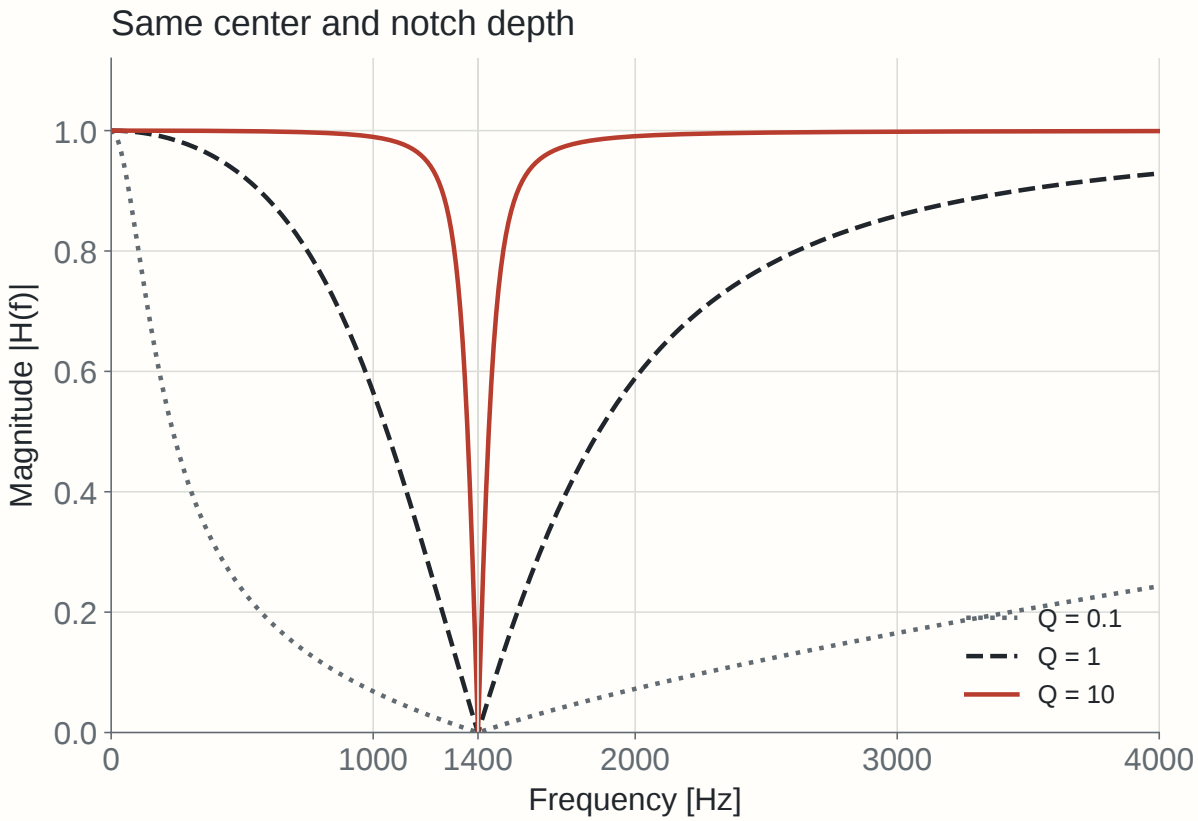
State the bandwidth convention: peaking EQ and digital filter APIs may use different definitions.



Notch: reject a narrow disturbance

- A notch is a narrow band-stop filter.
- All curves reject the same center: 1400 Hz.
- Increasing Q narrows the rejected band.
- In this ideal prototype, all reach zero at the center.
- Q does not independently specify notch depth.
- Frequencies outside the rejected band pass; nearby sound can still be affected.

Typical use: reduce a stable unwanted tone while preserving nearby sound.

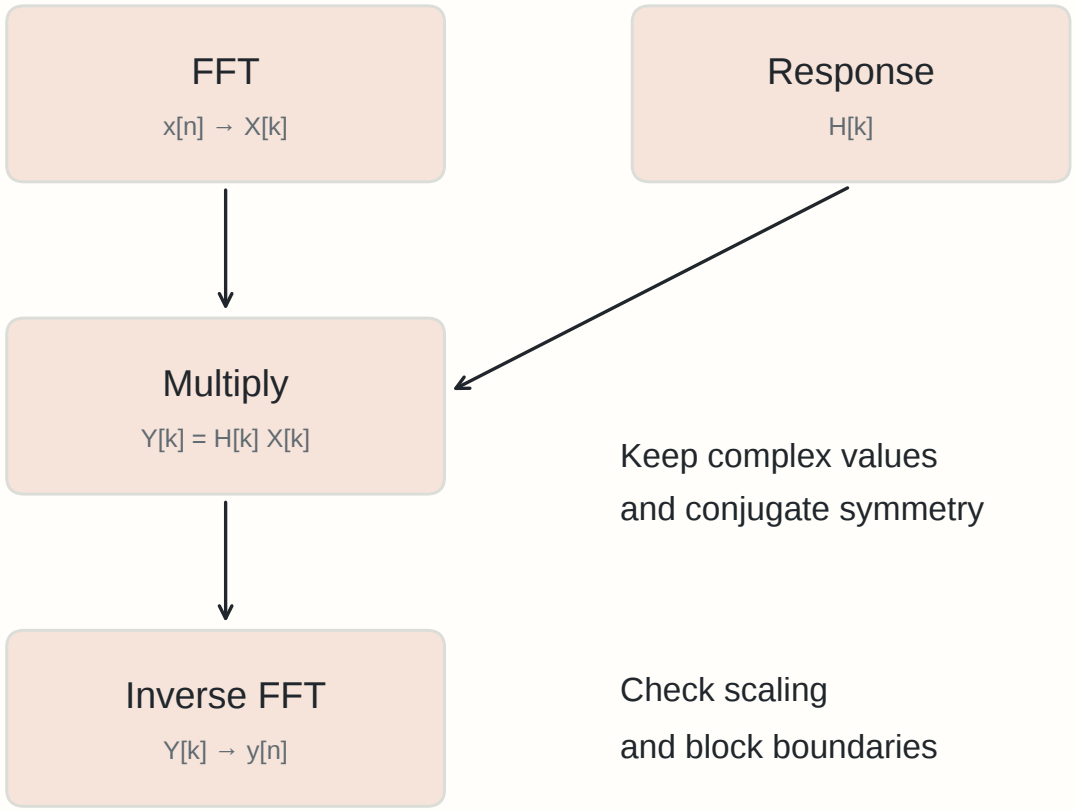


Filtering with the FFT

1. Transform a block: $X = \mathrm{FFT}(x)$.
2. Evaluate the filter response at the FFT bins.
3. Multiply: $Y[k] = H[k]X[k]$.
4. Reconstruct: $y = \mathrm{IFFT}(Y)$.

- Multiply complex values, retaining phase information.
- A real, nonnegative mask changes magnitude while preserving phase where nonzero.
- Check the library's inverse-transform normalization.

This basic block operation is circular filtering.

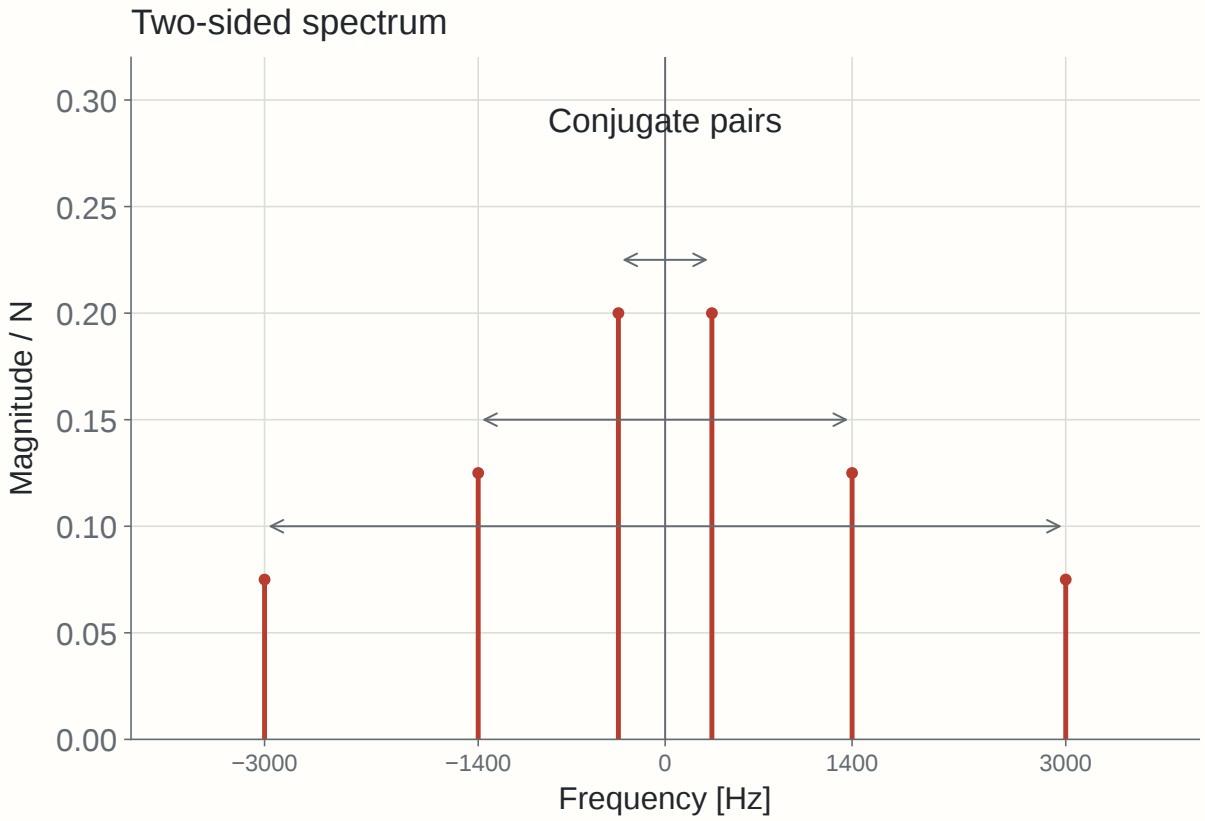


Real audio and spectral symmetry

For real-valued input:

$$X[N-k] = X[k]^*, \qquad k = 1, \ldots, N-1$$

- The second half represents negative frequencies.
- Preserve conjugate symmetry in H and Y to obtain real output.
- DC and, for even N, Nyquist coefficients are real.
- A real FFT stores only $\lfloor N/2 \rfloor + 1$ coefficients.
- For an inverse real FFT, pass the original length.



A one-sided display is not the full complex FFT array.

FFT filtering: padding, scaling and cropping

Zero-pad → FFT → symmetric complex-bin processing → IFFT → crop

- A radix-2 FFT requires $N = 2^m$; append zeroes to reach a supported length.
- With an unnormalized forward transform, reconstruction needs

$$y[n] = \frac{1}{N} \sum_{k=0}^{N-1} Y[k] e^{j2\pi kn/N}$$

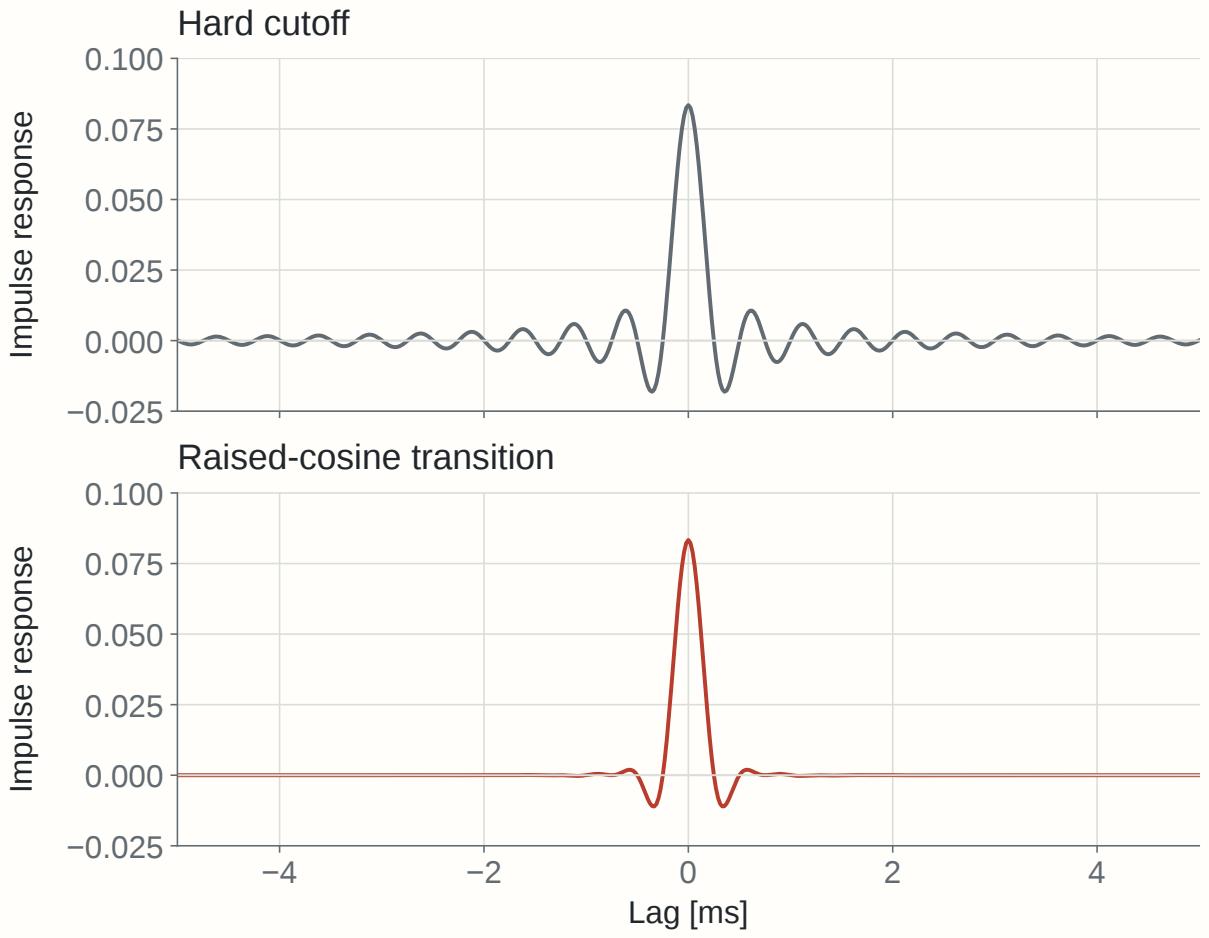
- Check whether the inverse routine already includes $1/N$ scaling.
- Padding changes bin spacing, $\Delta f = f_s/N$; it adds no measured information and does not by itself guarantee linear convolution.
- Cropping restores the original length, but can discard part of the filter response.
- An analysis window belongs to a spectrum measurement. Applying Hann to a whole processing buffer changes its amplitude envelope.

First check identity reconstruction, including padding and cropping.

Sharp cuts and ringing

- An abrupt spectral cutoff produces an extended, oscillatory impulse response.
- Around a transient, this can produce audible ringing.
- A smoother transition trades a less selective cutoff for reduced ringing.
- A zero-phase mask can spread energy before and after a transient.

Inspect the waveform as well as the spectrum.



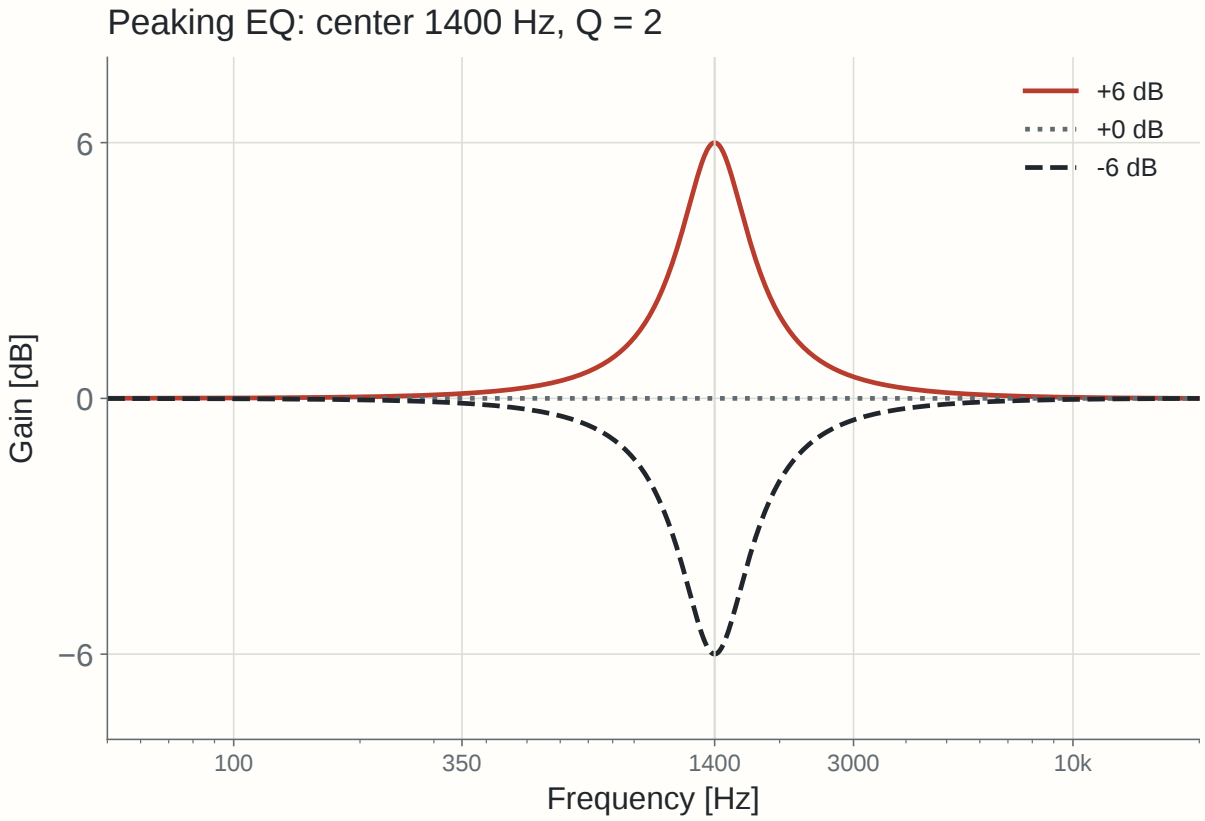
Peaking EQ: cut or boost a band

A parametric peaking band controls:

- Center frequency f_c.
- Q or bandwidth, with a stated convention.
- Gain: cut or boost in dB.

At 0 dB, the response is flat. Far from the affected band, gain approaches unity.

A graphic EQ usually fixes centers and widths; sliders control gain.



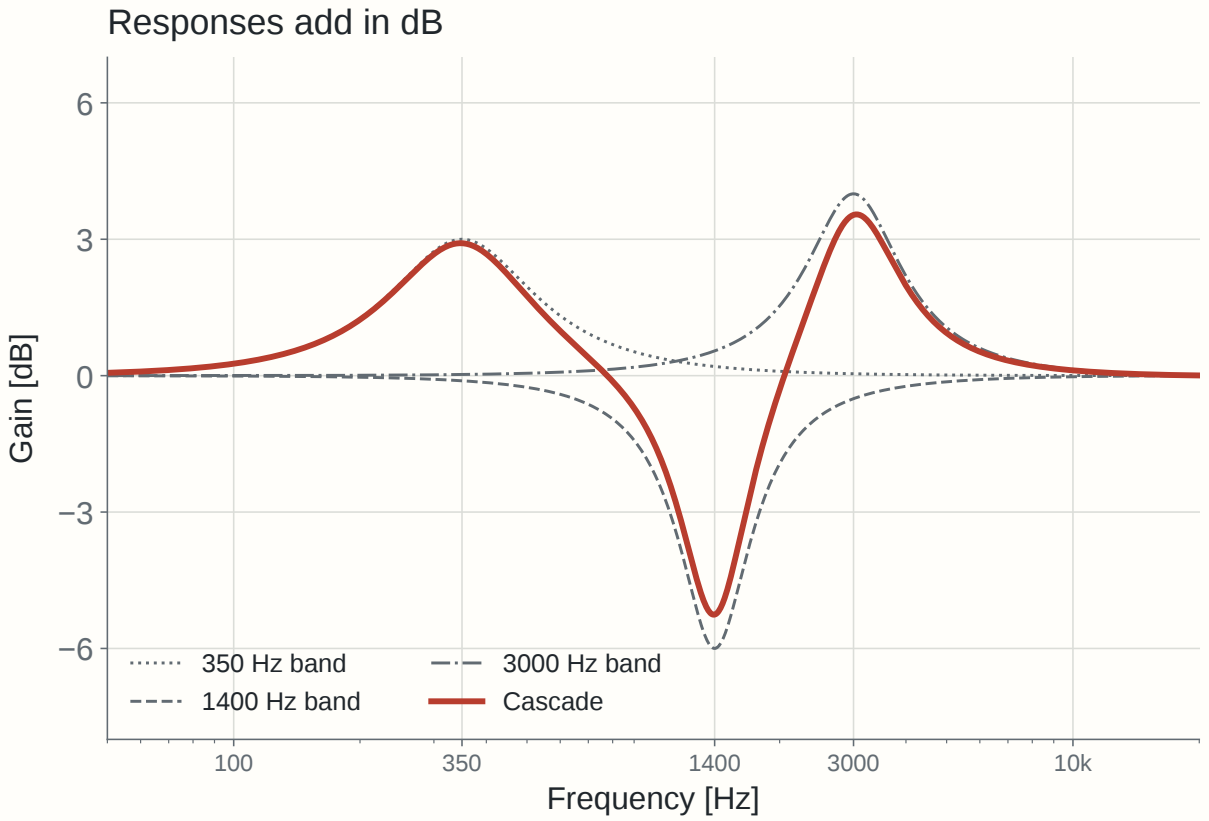
Combining EQ bands

Connect several filters in a cascade:

$$H_{\text{total}}(f) = \prod_i H_i(f)$$

Their magnitude responses add in dB.

- Each band affects a region, not just one frequency.
- Overlapping bands interact in the total response.
- A JSyn peaking-filter unit can process time samples directly; using it does not imply FFT processing.
- Check the selected API's gain units and Q convention.

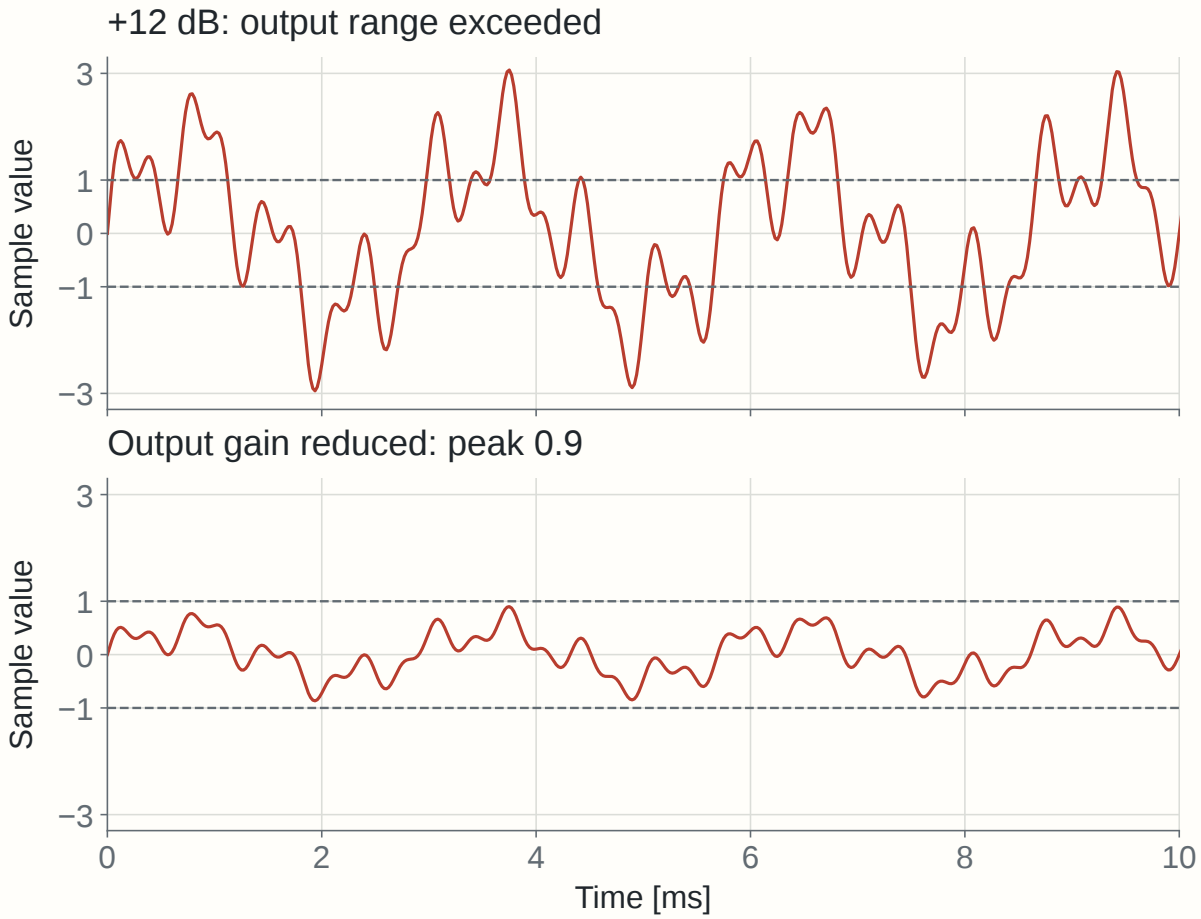


Gain, decibels, and headroom

For a positive amplitude multiplier g:

$$G_{\mathrm{dB}} = 20 \log_{10}(g), \qquad g = 10^{G_{\mathrm{dB}}/20}$$

- 0 dB: $g = 1$, unchanged amplitude.
- +6 dB: $g \approx 2$, approximately twice the amplitude.
- −6 dB: $g \approx 0.5$, approximately half the amplitude.
- dB describes a gain ratio; dBFS describes a signal level relative to digital full scale.
- Boosts can exceed the output range and cause clipping.
- Reduce input or output gain to leave headroom.

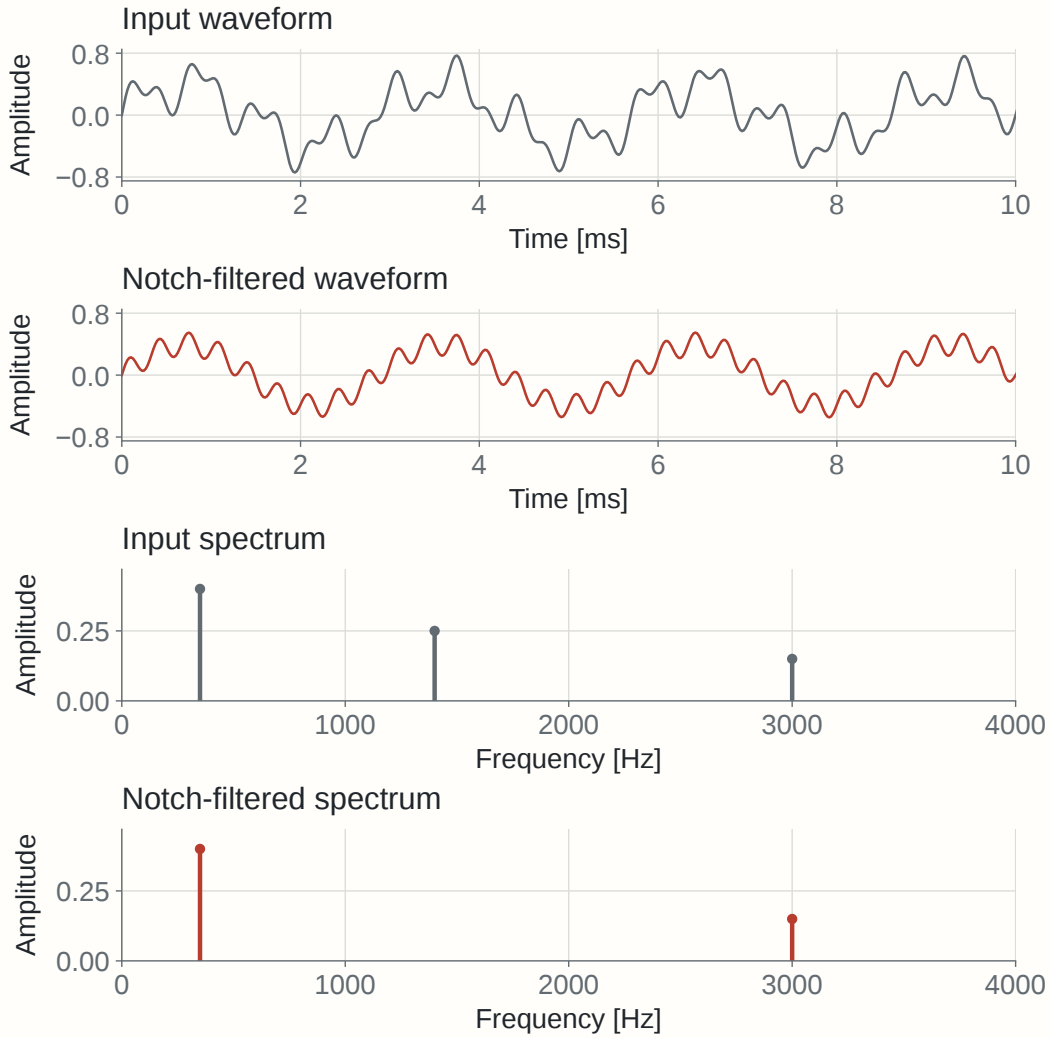


Checking the result

- Compare input and output on the same axes.
- Confirm which known components were retained or attenuated.
- Inspect transients and block joins for ringing or clicks.
- Listen before and after at comparable loudness.
- Check clipping, duration, and both stereo channels.

Here, the notch removes 1400 Hz and largely preserves 350 and 3000 Hz.

Successful filtering changes the intended components while preserving useful sound.



Appendix: spectrum calibration

For even N, choose $w[n] = 1$ or the periodic Hann window:

$$w[n] = \frac{1}{2}\left(1 - \cos\frac{2\pi n}{N}\right), \qquad X[k] = \mathrm{FFT}\{x[n]w[n]\}$$

With an unnormalized forward FFT, the one-sided peak amplitude is

$$P[k] = \frac{s_k|X[k]|}{\sum_{n=0}^{N-1} w[n]}, \qquad s_k = \begin{cases} 1, & k = 0 \text{ or } N/2 \\ 2, & 0 < k < N/2 \end{cases}$$

- Negative-frequency partners account for the factor of 2; DC and Nyquist stand alone.
- Dividing by the window sum compensates its coherent gain.
- Off-bin amplitude error and main-lobe widening remain.
- This is amplitude: power involves squared magnitude; PSD is power per Hz.

Amplitude levels and dBFS

dBFS expresses a digital level relative to full scale.

For a peak-amplitude spectrum with reference amplitude 1:

$$L_{\mathrm{dBFS}}[k] = 20 \log_{10}\left(\frac{P[k]}{1}\right)$$

Peak amplitude P	Level
1	0 dBFS
0.5	approximately -6.02 dBFS
0.1	-20 dBFS

- An isolated, bin-centered amplitude-1 sinusoid reads 0 dBFS after calibration.
- Response in dB is an output/input ratio; dBFS refers to a signal level.
- Display $20 \log_{10}(\max(P, 10^{-12}))$ to avoid $\log(0)$: the −240 dBFS floor is a plotting limit, not measurement precision.

From filter parameters to coefficients

For these second-order digital designs, define
$\omega_0 = 2\pi f_0 / f_s$, $c = \cos\omega_0$ and $\alpha = \sin\omega_0/(2Q)$.

Filter	b_0	b_1	b_2	a_0	a_1	a_2
Low-pass	$(1-c)/2$	$1-c$	$(1-c)/2$	$1+\alpha$	$-2c$	$1-\alpha$
High-pass	$(1+c)/2$	$-(1+c)$	$(1+c)/2$	$1+\alpha$	$-2c$	$1-\alpha$
Notch	1	$-2c$	1	$1+\alpha$	$-2c$	$1-\alpha$

These coefficients are unnormalized. Divide every coefficient by a_0:

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}, \qquad z = e^{j2\pi f/f_s}$$

- In this formula, b_i, a_i denote the normalized coefficients.
- Evaluate $|H(z)|$ for the amplitude response; with $Q = 1/\sqrt{2}$, LP/HP give approximately −3.01 dB at f_0.

Peaking EQ: gain and coefficient design

With $\omega_0 = 2\pi f_0 / f_s$, $c = \cos\omega_0$, $\alpha = \sin\omega_0/(2Q)$, let

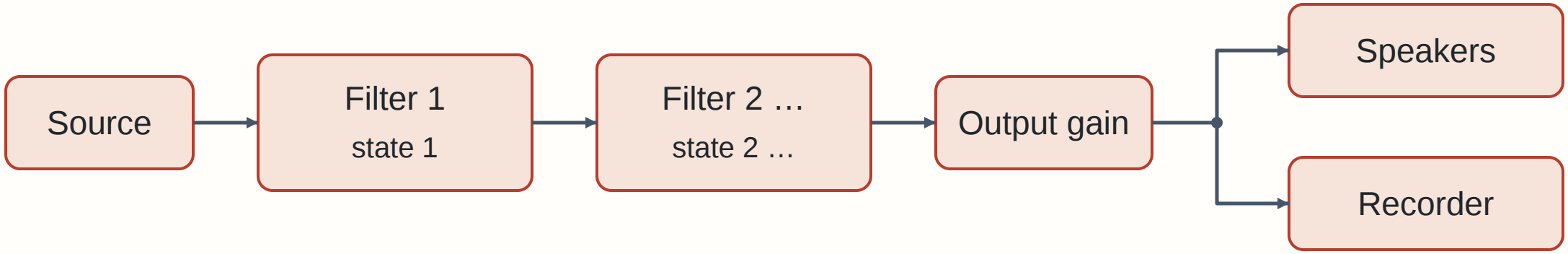
$$A = 10^{G_{\mathrm{dB}}/40}, \qquad g_{\mathrm{center}} = 10^{G_{\mathrm{dB}}/20} = A^2$$

b_0	b_1	b_2	a_0	a_1	a_2
$1 + \alpha A$	$-2c$	$1 - \alpha A$	$1 + \alpha/A$	$-2c$	$1 - \alpha/A$

- Divide these unnormalized coefficients by a_0.
- A is a coefficient-design intermediate; the center amplitude gain is A^2.
- At 0 dB, numerator and denominator match: the filter is an identity.
- Equal boost and cut at the same f_0 and Q have reciprocal responses; their cascade cancels within numerical tolerance.
- A control expecting a linear amplitude gain needs $10^{G_{\mathrm{dB}}/20}$.

Filter state, transients and signal routing

- Zero initial state means all remembered samples start at zero; switching on a signal can create a startup transient.
- A stable IIR filter can keep responding after the input stops: allow a decay tail.
- Preserve state across consecutive blocks; resetting it at every boundary can create discontinuities.
- Each stage in a cascade has independent state.
- Apply a common output gain before splitting to playback and recording.



The same processed signal should reach the listener and the recording.

Long recordings: block boundaries

FFT multiplication gives circular convolution: the end can wrap into the beginning.

For linear convolution with an L-sample block and an M-tap FIR filter:

$$N_{\mathrm{FFT}} \geq L + M - 1$$

- Zero-pad both arrays to the FFT length.
- Use overlap-add or overlap-save for long audio.
- Do not simply concatenate independently filtered blocks.
- Windowed spectral processing needs compatible analysis/synthesis windows and overlap.

